

Three-dimensional dynamic MR hysterosalpingography: a preliminary report.

PMID: 11519555 | Indexed in PubMed

The aim of this study was to evaluate the feasibility of three-dimensional dynamic MR hysterosalpingography (3D MR HSG) for visualization of the cavum uteri and demonstration of bilateral fallopian tube patency as an alternative to conventional hysterosalpingography. Five infertile female patients underwent 3D dynamic MR HSG prior to conventional hysterosalpingography. The MR protocol consisted of axial T1-weighted spin-echo (SE), axial/coronal T2-weighted fast SE (FSE), and 3D MR angiography sequences before, during, and after injection of a diluted gadolinium solution into the cavum uteri via a balloon catheter. Positioning of the catheter was feasible in all patients. In one patient the catheter slipped out during MRI and in one patient the catheter was placed far in the cavum uteri. In three patients catheter position was optimal at the level of the cervical canal. Evaluation of pelvic anatomy, myometrium, and ovaries was possible in all patients on the basis of T1-weighted SE and T2-weighted FSE. Three-dimensional visualization of the dilated cavum uteri was possible in four patients. In these four patients 3D MR HSG also proved bilateral fallopian tube patency which was confirmed in each patient by conventional hysterosalpingography. Three-dimensional MR HSG is feasible and further research should be done to determine if this technique can evolve into an alternative technique to conventional hysterosalpingography with the advantages of no radiation and additional visualization of the uterus wall and ovaries.

MR hysterosalpingography in a rabbit model.

PMID: 9132095 | Indexed in PubMed

Our objective was to evaluate the efficacy of MR imaging as compared with conventional hysterosalpingography (HSG) for the detection of fallopian tube patency after uterine injection of contrast material. Rabbit uterine horns ($n = 18$) were catheterized transvaginally. Five fallopian tubes were ligated and 11 were left unaltered. T1-weighted gradient-echo MR images were obtained before, during, and after injection of 1.0-3.0 mL of a dilute gadolinium-containing contrast agent. Corresponding conventional studies were performed with an equivalent volume of iohexol. Images were evaluated by two blinded readers. Observers agreed in all cases on the presence ($n = 11$) or absence ($n = 5$) of peritoneal spill with conventional HSG. Interpretation of MR HSG concurred with conventional HSG in 14 of 16 cases for each observer ($P > .05$). Reasons for misdiagnosis included small amounts of spill ($n = 2$), artifact ($n = 1$), and subtle spill between bowel loops ($n = 1$). Sensitivity and specificity for MR HSG were 95.5% and 70%, respectively, for the diagnosis of tubal patency/occlusions. No statistical difference was found between MR HSG and conventional HSG for the diagnosis of fallopian tube patency/obstruction. Potential advantages of MR HSG include no ionizing radiation, potentially diminished local contrast toxicity, superior visualization of uterine fibroids and endometriosis, and visualization of ovaries. We conclude that this technique warrants further study, including the use of a primate model to better simulate human anatomy.

A comparison of pain tolerance during X-ray hysterosalpingography and sonohysterosalpingography. Albunex Study Group.

PMID: 9067719 | Indexed in PubMed

The objective of this study was to determine patient perception of the level of discomfort during sonohysterosalpingography (sono-HSG) compared with standard X-ray hysterosalpingography (HSG). This was a prospective, operator-blinded, multicenter study using both numerical ranking and a visual analog scale to quantify discomfort. Sono-HSG utilizing Albunex was associated with a significantly lower mean intensity of pain than the instillation of X-ray contrast media. Sono-HSG is an alternative to X-ray HSG in the evaluation of the uterus and fallopian tubes and compares favorably with respect to patient discomfort.

Duplication of the fallopian tube.

PMID: 19562063 | Indexed in PubMed

Hysterosalpingography accurately delineates the uterine and tubal lumen, and hence is routinely performed for the evaluation of infertility. We observed a case of infertility where uterine cavity was normal but fallopian tubes were bifurcated at the ampullary region. Mullerian duct anomalies are reported in literature, but maldevelopment of fallopian tube in isolation is rare. This abnormality can present as infertility, ectopic pregnancy, in association with urinary tract anomalies or as failure of sterilisation method.

[Radiologic versus ultrasound fallopian tube imaging. Painfulness of the examination and diagnostic reliability of hysterosalpingography and hysterosalpingo-contrast-ultrasonography with echovist 200].

PMID: 9173524 | Indexed in PubMed

Evaluation of tubal patency is usually assessed with hysterosalpingography (HSG) or laparoscopy including chromopertubation. Sonographical visualisation with Echovist 200 (hysterosalpingo-contrast sonography-HyCoSy) provides a new noninvasive tool. Therefore we conducted a prospective controlled study to compare sonographic and radiological evaluation of the fallopian tube. Main test parameters were accuracy of both methods and patient discomfort. 50 patients were enrolled in this study. All patients were examined by both techniques; the sequence was randomly chosen. The results of HSG and HyCoSy were compared. Patient discomfort was assessed with a standardised questionnaire using a visual analog scale (0-10). Diagnosis of tubal patency identifying proximal or distal blockage was the primary end point using HSG as standard technique. Proximal and distal patency by HSG was sonographically confirmed in 82.9% (63/76) and 82.1% (46/56) tubes respectively. If HSG revealed proximal or distal occlusion, identical results were obtained in 91.7% (22/24) or 60% (12/20) by HyCoSy. No significant differences were found in patient discomfort. However a significant correlation was demonstrated between tubal patency and discomfort. The lowest score was obtained in patients with open tubes (4.6) increased in patients with distal occlusion (6.0) and reached a maximum with proximal pathology (8.7). Compared to conventional HSG, HyCoSy provides a highly efficient evaluation of tubal pathology and can be successfully used as a noninvasive screening method.

Mutational spectrum of CFTR in cystic fibrosis patients with gastrointestinal and

hepatobiliary manifestations.

PMID: 38662334 | Indexed in PubMed

Cystic fibrosis (CF) is a rare and debilitating autosomal recessive disorder. It hampers the normal function of various organs and causes severe damage to the lungs, and digestive system leading to recurring pneumonia. Cf also affects reproductive health eventually may cause infertility. The disease manifests due to genetic aberrations in the cystic fibrosis transmembrane conductance regulator (CFTR) gene. This study aimed to screen for CFTR gene variants in Pakistani CF patients representing variable phenotypes. Clinical exome and Sanger sequencing were performed after clinical characterization of 25 suspected cases of CF (CF1-CF25). ACMG guidelines were followed to interpret the clinical significance of the identified variants. Clinical investigations revealed common phenotypes such as pancreatic insufficiency, chest infections, chronic liver and lung diseases. Some patients also displayed symptoms like gastroesophageal reflux disease (GERD), neonatal cholestasis, acrodermatitis, diabetes mellitus, and abnormal malabsorptive stools. Genetic analysis of the 25 CF patients identified deleterious variants in the CFTR gene. Notably, 12% of patients showed compound heterozygous variants, while 88% had homozygous variants. The most prevalent variant was p. (Met1Thr or Met1?) at 24%, previously not reported in the Pakistani population. The second most common variant was p. (Phe508del) at 16%. Other variants, including p. (Leu218*), p. (Tyr569Asp), p. (Glu585Ter), and p. (Arg1162*) were also identified in the present study. Genetic analysis of one of the present patients showed a pathogenic variant in G6PD in addition to CFTR. The study reports novel and reported variants in the CFTR gene in CF patients in Pakistani population having distinct phenotypes. It also emphasizes screening suspected Pakistani CF patients for the p. (Met1Thr) variant because of its increased observance and prevalence in the study. Moreover, the findings also signify searching for additional pathogenic variants in the genome of CF patients, which may modify the phenotypes. The findings contribute valuable information for the diagnosis, genetic counseling, and potential therapeutic strategies for CF patients in Pakistan.

Mutational Spectrum of the CFTR Gene in the Kazakhstan Population.

PMID: 35273129 | Indexed in PubMed

To study the frequency and spectrum of CFTR gene variants in different ethnic groups of Kazakhstan. We reviewed the records of 58 patients with cystic fibrosis. All the patients underwent molecular genetic analysis to reveal genotype-phenotype correlations. The median (IQR) age of the patients was 5.4 year (7 months, 18 year); 40% were diagnosed at the age of 5-10 year. The study identified 28 specific variants: p.Phe508del, the variant most common in the European population, was detected in 30 patients (51.7%). Variants other than p.Phe508del were revealed in 31% (21 patients). We found a number of specific variants characteristic of the Kazakhstani population. A pronounced regression of disease symptoms was detected in patients with mild mutations; whereas in patients with severe mutations, therapy produced very little effect.

Accounting for population structure in genetic studies of cystic fibrosis.

PMID: 35647563 | Indexed in PubMed

CFTR F508del (c.1521_1523delCTT, p.Phe508delPhe) is the most common pathogenic allele underlying

cystic fibrosis (CF), and its frequency varies in a geographic cline across Europe. We hypothesized that genetic variation associated with this cline is overrepresented in a large cohort ($N > 5,000$) of persons with CF who underwent whole-genome sequencing and that this pattern could result in spurious associations between variants correlated with both the F508del genotype and CF-related outcomes. Using principal-component (PC) analyses, we showed that variation in the CFTR region disproportionately contributes to a PC explaining a relatively high proportion of genetic variance. Variation near CFTR was correlated with population structure among persons with CF, and this correlation was driven by a subset of the sample inferred to have European ancestry. We performed genome-wide association studies comparing persons with CF with one versus two copies of the F508del allele; this allowed us to identify genetic variation associated with the F508del allele and to determine that standard PC-adjustment strategies eliminated the significant association signals. Our results suggest that PC adjustment can adequately prevent spurious associations between genetic variants and CF-related traits and are therefore effective tools to control for population structure even when population structure is confounded with disease severity and a common pathogenic variant.

National Newborn Screening for cystic fibrosis in the Republic of Ireland: genetic data from the first 6.5 years.

PMID: 32483343 | Indexed in PubMed

Cystic fibrosis (CF) is the most common life-limiting autosomal recessive disease in the Republic of Ireland (ROI), with a previously quoted incidence of 1 in 1353 and carrier rate of 1 in 19. The National Newborn Screening (NBS) for CF was incorporated in July 2011 in the ROI. A cut-off point of the top 1% Immunoreactive Trypsinogen (IRT) was taken as an indication for 38 CFTR variant panel to maximise identification of affected CF cases and to minimise detection of carriers. All neonates from July 2011 to Dec 2017 with an elevated IRT on NBS were tested with 38 CFTR mutation panel and included. Clinical and laboratory database were analysed. In the first 6.5 years a total of 5,053 newborns (1.16% of total births) were screened with 38 CFTR panel. 170 CF affected cases, 320 unaffected carriers, 32 CF Screening Positive Inconclusive Diagnosis (CFSPID) were identified. There was one missed diagnosis. The most common disease-causing variant was c.1521_1523delCTT (p.(Phe508del)) followed by c.1652G>A (p.(Gly551Asp)). 95 out of 170 (55%) affected newborns were homozygous for c.1521_1523delCTT (p.(Phe08del)) and 25 (15%) carried at least one copy of c.1652G>A (p.(Gly551Asp)). Hence, 70% of affected newborns were eligible for CFTR modulator treatment. The NBS programme has identified almost triple the number of affected newborn with c.1652G>A (p.(Gly551Asp)) than previously quoted figures and identified less than 50% of carriers than predicted. The revised incidence and carrier frequency of CF in the ROI is 1 in 2570 and 1 in 25, respectively.

A Rare Cystic Fibrosis Transmembrane Conductance Regulator (CFTR) Mutation Associated With Typical Cystic Fibrosis in an Arab Child.

PMID: 33786233 | Indexed in PubMed

Cystic fibrosis (CF) is a progressive genetic disorder, inherited by the autosomal recessive mode of inheritance and more frequently seen in the Caucasian population with a carrier rate of 1:29 in Caucasian-Americans. Over 1800 cystic fibrosis transmembrane conductance regulator (CFTR) gene

mutations have been identified so far and the delta F 508 del mutation is the most common mutation. Gene sequencing and deletion/duplication analysis can detect mutations in 99% of people with a clinical diagnosis of CF. However, diagnostic testing can be challenging, as screening tests may be inconclusive and the routine gene mutation panel analysis may be negative due to some rare or undocumented mutations. We report a case of a two-year-old boy of Palestinian-Lebanese descent, with a history of raised immunoreactive trypsin test (IRT), positive sweat test, and phenotypical CF manifestations, found to have rare CF apparent homozygous CFTR (NM_000492.3) variant, c.3623del (p.Gly1208AlafsX3). In our case, genetic testing for 139 mutations done in Germany could not identify any defect. Only CFTR gene sequencing identified the above pathogenic variant. This reinforces the practice for a broad range of CFTR mutation analyses to detect ethnic-specific rare variants. This is the second case of this particular genetic mutation identified and the first to be reported in detail.
